Mesa Example Models Audit Report

Introduction

This document presents a structured audit of example models available in the Mesa repository. The audit was conducted to evaluate the current functionality, compatibility, and usability of these examples when executed in a modern Mesa environment.

Example models play an important role in the Mesa ecosystem. They serve as learning resources for new users, reference implementations for common agent-based modeling patterns, and demonstration tools for Mesa's core features such as scheduling, agent interactions, data collection, and visualization. Ensuring that these examples run correctly and reflect the current Mesa APIs is therefore important for both educational and development purposes.

As part of this audit, a total of **20 Mesa example models** were tested locally. Each example was reviewed by installing its required dependencies, examining the provided instructions, and attempting to run the model using the recommended execution commands (such as `solara run app.py`, `mesa runserver`, or notebook execution).

For every example model included in this report, the following aspects were evaluated:

* Clarity and completeness of installation instructions
* Required dependencies and environment setup
* Execution success or failure

* Runtime errors or warnings encountered

* Compatibility with the current Mesa visualization system and APIs

* Categorization of identified issues (e.g., legacy visualization, outdated CLI usage, packaging issues, dependency specification problems, or documentation gaps)

Based on these evaluations, each example has been classified into categories such as **stable**, **documentation issue**, **legacy visualization**, **API incompatibility**, or **dependency specification issue**. This classification aims to help identify which examples currently work as expected and which may require maintenance or modernization.

The findings summarized in this document are intended to assist Mesa maintainers and contributors in understanding the current state of the example models and identifying potential areas for improvement, including modernization of legacy examples and clarification of installation instructions.

Audit conducted by:

Tushar Mahato

Scope of audit: 20 Mesa example models

Environment: Local execution using the latest available Mesa setup at the time of testing

1. aco_tsp

Install instructions: Clear

Dependencies needed:

mesa

solara

networkx

numpy

Run command:

solara run app.py

Result: Runs successfully (after installing missing dependencies)

Issue:

Occasional runtime error:

ValueError: Total of weights must be finite

Category:

Mostly stable / occasional runtime instability

2. boltzmann_wealth_model_network

Install instructions: Clear

Dependencies needed:

mesa[rec]

Run command:

solara run app.py

Result: Runs successfully

Issue: None

Category:

Stable

3. caching_and_replay

Install instructions: Clear

Dependencies needed:

jupyter

mesa

Run command:

python run.py

mesa runserver

Result: Did not run

Issue:

AttributeError: module 'mesa' has no attribute 'visualization'

Category:

Legacy visualization / outdated API

4. charts

Install instructions: Good

Dependencies needed:

itertools

mesa~=2.0

numpy

pandas

Run command:

mesa runserver

Result: Did not run

Issue:

Deprecated Mesa visualization (CanvasGrid).

Category:

Outdated example

5. color_patches

Install instructions: Not mentioned

Dependencies needed:

mesa[viz]>=3.0

networkx

Run command:

mesa runserver

Result: Did not run

Issue:

mesa command not recognized

Category:

Legacy CLI

6. conways_game_of_life_fast

Install instructions: Clear

Dependencies needed:

mesa

numpy

scipy

Run command:

solara run app.py

Result: Did not run

Issue:

ValueError: Unknown space type: <class 'NoneType'>

Category:

Visualization compatibility issue

7. deffuant_weisbuch

Install instructions: Clear

Dependencies needed:

mesa[rec]

Run command:

solara run app.py

Result: Runs successfully

Issue:

Matplotlib warning about too many open figures.

Category:

Stable (minor warning)

8. dining_philosophers

Install instructions: Clear

Dependencies needed:

mesa

solara

networkx

matplotlib

Run command:

solara run app.py

Result: Runs successfully

Issue: None

Category:

Stable

9. el_farol

Install instructions: Unclear

Dependencies needed:

jupyter

matplotlib

mesa

numpy

seaborn

Run command:

Notebook execution

Result: Could not run

Issue:

Instructions unclear.

Category:

Documentation issue

10. emperor_dilemma

Install instructions: Clear

Dependencies needed:

mesa

matplotlib

solara

Run command:

solara run app.py

Result: Did not run

Issue:

ImportError: attempted relative import with no known parent package

Category:

Packaging/import issue

11. hex_ant

Install instructions: Clear

Dependencies needed:

mesa[rec]

Run command:

solara run app.py

Result: Did not run

Issue:

ImportError: attempted relative import with no known parent package

Category:

Packaging/import issue

12. hex_snowflake

Install instructions: Clear

Dependencies needed:

mesa

Run command:

mesa runserver

Result: Did not run

Issue:

Legacy CLI.

Category:

Legacy visualization

13. hotelling_law

Install instructions: Clear

Dependencies needed:

mesa[rec]

Run command:

solara run app.py

Result: Runs successfully

Issue: None

Category:

Stable

14. humanitarian_aid_distribution

Install instructions: Partial

Dependencies needed:

mesa

solara

matplotlib

numpy

Run command:

solara run app.py

Result: Runs successfully

Issue:

Missing dependency instructions.

Category:

Documentation issue

15. rumor_mill

Install instructions: Clear

Dependencies needed:

mesa[rec]

Run command:

solara run app.py

Result: Runs successfully

Issue: None

Category:

Stable

16. shape_model

Install instructions: Clear

Dependencies needed:

mesa

Run command:

mesa runserver

Result: Did not run

Issue:

Legacy Mesa CLI removed.

Category:

Legacy visualization

17. termites

Install instructions: Partial

Dependencies needed:

mesa

solara

numpy

matplotlib

Run command:

solara run app.py

Result: Did not run

Issue:

TypeError: HasPropertyLayers.add_property_layer() takes 2 positional arguments but 3 were given

Category:

API incompatibility

18. virus_antibodies

Install instructions: Partial

Dependencies needed:

mesa

solara

matplotlib

numpy

Run command:

solara run app.py

Result: Runs successfully

Issue:

Incomplete installation instructions.

Category:

Documentation issue

19. warehouse

Install instructions: Clear

Dependencies needed:

mesa[rec]>=3

Run command:

solara run app.py

Result: Runs successfully

Issue: None

Category:

Stable

20. forest_fire

Install instructions: Clear

Dependencies needed:

mesa

solara

networkx

Run command:

solara run app.py

Result: Runs successfully

Issue:

networkx dependency missing in requirements.

Category:

Dependency specification issue